Do Register Artifacts Survive Parameter-Efficient Fine-Tuning?
Training-Free Registers for Dense Prediction with LoRA-Adapted Vision Transformers

John Roth · EN.705.744 Deep Learning using Transformers, Fall 2026 · Research Project Proposal

Outline (Research Project Rubric)

1. **Introduction & Goals.** Register artifacts in ViTs, their cost for dense prediction, and the gap: no prior study under PEFT.
2. **Hypotheses & Method.** Three falsifiable hypotheses; attention, LoRA and register math; block diagram (Fig. 1); ADE20K prototypes.
3. **Related Work.** Registers [1,2], DINOv2/v3 [3,4], CLIP [5], LoRA and vision PEFT [6], segmentation heads [7], maritime segmentation [8].
4. **Application.** 3×3 factorial (adaptation $\times$ register condition) on three datasets, sweeps, 3 seeds, statistics, throughput.
5. **Conclusions.** The hypotheses that held, a PEFT recipe for DINOv2/CLIP dense tasks, and the reason for the study.

Problem, Candidate Solution, and Hypotheses

Problem. Large pretrained ViTs (DINOv2, CLIP) develop a sparse set of high-norm "outlier" patch tokens in low-information regions (sky, water, walls). The model uses them as global memory, which corrupts attention and feature maps and lowers segmentation quality [1]. Darcet et al. [1] remove them with extra *register* tokens, which requires retraining the foundation model. Jiang et al. [2] traced the artifacts to a few "register neurons" in one MLP layer and removed them without training by redirecting those activations into an appended, untrained token at inference (*test-time registers*). Practitioners adapt these backbones with LoRA, and no published work measures whether (a) LoRA removes the artifacts by itself, (b) test-time registers still help once LoRA is applied, or (c) the effect grows for images dominated by low-information regions.

Candidate solution. *Register-aware PEFT*: (i) identify the register-neuron set once on the frozen backbone; (ii) insert test-time register tokens; (iii) train low-rank adapters on the attention projections and a light segmentation head, nothing else. The method adds no learned parameters over plain LoRA and needs no access to pretraining. **Research hypotheses:**

- **H1 (persistence).** LoRA fine-tuning leaves the outlier tokens in place: the fraction of high-norm patch tokens after LoRA stays within $\pm 20\%$ of the frozen model, and vanilla LoRA scores below a backbone trained with registers on mIoU.
- **H2 (recovery).** Test-time registers + LoRA close at least 50% of the mIoU gap between vanilla LoRA and the trained-register backbone, at equal parameter count.
- **H3 (scaling with low-information area).** The gain grows with the fraction of homogeneous background. It peaks on maritime imagery (LaRS, where sky and water dominate) and is larger for CLIP than for DINOv2.

Applied Machine Learning Method and Datasets

Backbones and math. Each ViT block computes $\mathrm{Attn}(X) = \mathrm{softmax}\big(XW_Q W_K^\top X^\top/\sqrt{d}\big)XW_V$. LoRA [6] freezes $W \in \mathbb{R}^{d\times k}$ and learns $W' = W + \frac{\alpha}{r}BA$ with $B \in \mathbb{R}^{d\times r}, A \in \mathbb{R}^{r\times k}, r \ll d$, applied to W_Q, W_V. We define outlier tokens as $\{i : \|x_i^{(\ell)}\|_2 > \tau\}$ and set τ from the norm distribution of a calibration set. Following [2], we take the register neurons $\mathcal{N}$ to be the MLP units whose activation magnitude on outlier tokens exceeds a high quantile; at layer ℓ we set $h_i[\mathcal{N}] \leftarrow 0$ for all patch tokens i and write the removed activations to m appended register tokens, which hold the global information away from the patch features. Backbones: DINOv2 ViT-S/14 and ViT-B/14 with and without trained registers (the upper bound) [3]; CLIP ViT-B/16 [5]; DINOv3 ViT-B/16 [4] as a stretch goal.

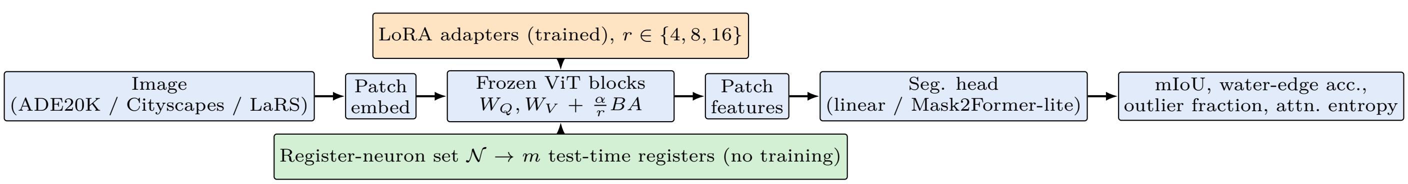

Fig. 1. Register-aware PEFT. Orange: the only trained backbone parameters. Green: the training-free intervention from [2]. Ablation arms drop orange, green, or both, and we compare each against the trained-register backbone.

Experimental design. We run a factorial over adaptation {frozen, LoRA, full fine-tune} $\times$ register condition {none, test-time, trained} on each backbone, sweep rank r, target modules, number of registers $m \in \{1, 4, 8\}$, LoRA layer subset and learning rate, and report mean±std over 3 seeds per cell with paired tests. Heads: a linear probe, which isolates feature quality, and a light Mask2Former-style head [7].

Datasets and metrics. ADE20K (150 classes, 20k train) [9] and Cityscapes (19 classes, 2,975 train) [10] are the standard benchmarks; LaRS [8] ($\approx$4k per-pixel labelled maritime frames, obstacle/water/sky) is the low-information stress test for H3, scored with its water-edge accuracy and obstacle F1 as well as mIoU. Diagnostics: fraction of tokens above τ, attention entropy of the CLS/register tokens, and images/s. Compute: one 24 GB GPU; with backbones frozen, all cells fit in $\approx$8 GPU-weeks. **Timeline:** weeks 1–2 reproduce [2] and build the outlier diagnostics; weeks 3–6 ADE20K factorial (draft due 10/19); weeks 7–10 Cityscapes and LaRS, sweeps, seeds; weeks 11–13 analysis, CLIP, writing. Venues: MaCVi workshop at WACV 2027 for the LaRS slice, then a CVPR 2027 workshop or WACV 2028 for the full study.

References

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